Reg. No. | | | | | | | | | | | [18MTSE231]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

DESIGN CONCEPTS OF SUB STRUCTURES

Time: 03 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question 1&2 are compulsory.**
ii. Missing data, if any, may be suitably assumed.

1. a. What are the general criteria to be adopted for selecting the type of foundation? **10**

b. Explain the Standard Penetration Test with its advantage and disadvantage. **10**

2. a. What are the components of settlement? Explain them briefly. **10**

b. What will be the maximum safe bearing load which can be carried by a footing for FOS=3 against complete shear failure. The square footing is 2.5mx2.5m built in homogeneous bed of sand with unit weight 16KN/m^3 in loose state, the angle of shearing resistance is 25^0 and the depth of the base of the foundation is 1.5m. **10**

3. a. Explain briefly about the factors affecting bearing capacity of foundation. **10**

b. With neat sketch explain the types of raft foundation. **10**

OR

4. a. What are the types of combined footing? Explain any two of them. **10**

b. A square footing of 2.5m X 2.5m is built in a homogeneous bed of sand of unit weight 20 KN/m^3 and having an angle of shearing resistance 36^0. The depth of base of footing is 1.5m below the ground surface. Calculate the safe load that can be carried by a footing with a FOS =3 with respect to complete shear failure, use Terzaghi's equation. **10**

5. a. What are the types of deep foundation? Explain with neat sketch. **10**

b. Explain ultimate bearing capacity of different types of piles in different soil conditions. **10**

OR

6. a. What are the types of Piles and explain them briefly? **10**

b. In a 16 pile group, pile dia 45cm and c/c spacing is 1.5m of C=15 kpa. Determine whether failure would occur individually or as a group, neglect end bearing at the tip of the pile. All the piles are 10m long, take α =0.7. **10**

7. a. Explain advantages and disadvantages of caisson foundation. What are the uses of caisson foundation? **10**

b. What are the precautions to be taken for avoiding and remedial measure for rectifying the tilts and shifts? **10**

OR

8. a. Explain forces acting on tower foundations. **10**

b. Discuss the design of foundation of transmission towers in different types of soils. **10**

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